

SatyaScan System Architecture

Technical Report on Text Analysis, Image Forensics & URL Verification

Generated: 8/8/2026 | Multi-Modal Digital Authenticity Engine

1. Executive Overview

Multi-Modal Truth & Digital Authenticity Engine

SatyaScan is an enterprise-grade, multi-modal fact-checking and digital authenticity verification platform. It addresses the growing challenge of misinformation, cheapfakes, AI hallucinations, and manipulated visual media through three tightly integrated, specialized analysis pipelines:

System Architecture Breakdown

- Text Analysis Pipeline: Multi-source search orchestration, Named Entity Recognition, dual-tier trust ranking, and strictly grounded Gemini reasoning.
- Image Forensics Pipeline: Deep EXIF parsing, camera/GPS verification, digital manipulation signatures, and multimodal Gemini 2.5 Flash Vision.
- URL & Live Webpage Pipeline: Heuristic URL categorization (Official, Reference, Opinion, News), HTML text extraction, and live Chrome extension in-browser scanning.

KEY TAKEAWAY: Strict Zero-Hallucination policy: The reasoning engine evaluates ONLY verified retrieved evidence.

Core Technology Stack

- AI & Multimodal Reasoning: Google Gemini 2.5 Flash API (Structured JSON output, Temperature 0.1)
- Verification Search Engines: Google Fact Check Tools API + Tavily Search API (Hybrid Open & Domain-Restricted passes)
- Digital Image Forensics: exifr (EXIF, IPTC, XMP, TIFF parsing) + Forensic software signature detection
- Scraping & Entity Extraction: Custom Rule-Based Named Entity Recognizer (NER), Axios HTTP extractor
- Backend & Storage: Node.js, Express.js, MongoDB (Mongoose), Winston Logger, Rate Limiter
- Client Integrations: React 18 Web Application (Vite, TailwindCSS) + Chrome Extension (Manifest V3 Service Worker)

2. Text Analysis Pipeline (Claim Verification)

Comprehensive fact-checking workflow from claim ingestion to structured verdict

The Text Analysis pipeline processes arbitrary user-entered claims or highlighted text selections. It follows an 8-stage sequence designed to maximize recall from reliable fact-checkers while preventing AI hallucination.

1 24-Hour Normalized Claim Caching

Before hitting external APIs, incoming claims are normalized (lowercased, stripped of non-alphanumeric/Devanagari characters) and queried against MongoDB. Exact matches within 24 hours in the requested language return cached verdicts instantly in 0.0s.

Implementation: analyzeController.js (Lines 32-78), Check.js

2 Named Entity Recognition (NER) & Language Detection

A custom entity extractor identifies people, organizations, locations, events, and dates using curated Indian and international vocabularies. It also detects language distribution (English, Hindi, or Hinglish) to configure localized response prompts.

Implementation: entityExtractor.js (analyzeClaimForSearch, detectLanguage)

3 Diversified Entity-Aware Query Generation

Generates 5-8 diversified search query vectors: the raw claim, fact-checking queries ("<claim> fact check"), official queries ("<claim> official statement"), and entity-specific news vectors ("<entity> news"). Capped to top 5 queries for performance.

Implementation: textVerificationService.js (Lines 91-116)

4 Parallel Dual-Engine Evidence Retrieval

Executes three parallel asynchronous requests:

- Google Fact Check Tools API: Direct query against accredited fact-checkers (Snopes, AltNews, BoomLive, Reuters, Factly).
- Tavily Hybrid Search: Simultaneous open search and trusted-domain restricted search across Tier 1 news sources.
- Official Government Search: Targeted query across gov.in, nic.in, who.int, and un.org portals.

Implementation: factCheckService.js, tavilyService.js (searchMultiple)

5 Multi-Tier Evidence Ranking & Deduplication

Combines all retrieved sources, removes duplicates by URL, and ranks via a strict 7-tier custom priority order:

- Official Government Domains (.gov, .nic.in, who.int) !
- Google Fact Check hits !
- Wire Services (Reuters, AP, AFP) !
- International Broadcasters (BBC) !
- Premium National News (The Hindu, Indian Express, NDTV) !
- Source Tier (1-4) !
- Relevance Score. Top 8 sources are packaged for the AI.

Implementation: textVerificationService.js (Lines 145-205), helpers.js

6 Strictly Grounded Gemini Reasoning

Sends the claim and top 8 ranked evidence snippets to Gemini Flash. The prompt enforces a strict zero-hallucination constraint: the model must reason solely over retrieved snippets without using internal memory.

Implementation: prompts/textVerification.js, geminiService.js (analyzeText)

7 Multi-Factor Trust Score & Metric Computation

Calculates overall Trust Score (0-100), AI Likelihood score, source credibility index, evidence distribution breakdown (Supporting vs. Contradicting vs. Neutral counts), and 6-dimension confidence breakdown (Evidence Quality, Independent Sources, Official Confirmation, Recent Reporting, Contradictions, AI Consistency).

Implementation: textVerificationService.js (Lines 240-280)

3. Image Analysis Pipeline (Digital Forensics & Deepfake Detection)

Multi-stage forensic pipeline inspecting file metadata, manipulation signatures, and visual AI artifacts

Images are routed to a dedicated digital forensics workflow. Unlike text, image analysis does NOT execute web searches; instead, it combines deep metadata inspection with multimodal computer vision analysis via Gemini 2.5 Flash Vision.

1 Buffer Validation & Ingestion Control

Validates multipart image stream, verifies MIME type (JPEG, PNG, WebP), and enforces a 10MB file size limit to prevent memory exhaustion.

Implementation: imageVerificationService.js (Lines 19-25)

2 Deep EXIF & Photographic Metadata Extraction

Uses exifr to extract comprehensive metadata tags: Camera Make/Model (e.g., Canon, Nikon, Apple iPhone), Software tag, Original DateTime, Image Dimensions, GPS Coordinates (Latitude/Longitude), ISO, Exposure Time, F-Number (Aperture), and Focal Length.

Implementation: utils/exifParser.js (extractExifData)

3 Software Signature & Metadata Integrity Analysis

Inspects metadata for forensic signals:

- Editing Software: Detects Photoshop, GIMP, Lightroom, Snapseed, PicsArt, Canva (flags MODIFIED integrity).
- AI Generators: Scans for Midjourney, DALL-E, Stable Diffusion, ComfyUI, Automatic1111, Firefly.
- Integrity States: INTACT (complete camera/GPS/timestamp), MODIFIED (edited), STRIPPED (metadata deleted/cleaned), or UNKNOWN.

Implementation: imageVerificationService.js (analyzeMetadata)

4 Multimodal Gemini 2.5 Flash Vision Analysis

Encodes the image buffer as base64 and passes it alongside the extracted EXIF summary to Gemini Vision (temperature 0.2). The model inspects:

- AI Generation Artifacts: Unnatural hair/skin smoothing, plastic sheen, asymmetric eye reflections, irregular fine detail patterns (fingers, teeth, background text).
- Deepfake Anomalies: Face/body boundary mismatches, lighting/shadow direction conflicts, warping artifacts.
- Digital Tampering: Clone/stamp traces, inconsistent compression noise, compositing edge seams, inpainting artifacts.

Implementation: prompts/imageVerification.js, geminiService.js (analyzeImage)

5 Probabilistic Forensics Verdict & Reporting

Generates independent probability scores (AI Probability 0-100%, Deepfake Probability 0-100%, Manipulation Probability 0-100%) and categorizes the image into definitive verdicts:

AUTHENTIC | LIKELY_AUTHENTIC | AI_GENERATED | LIKELY_AI_GENERATED | DEEPFAKE | MANIPULATED | INCONCLUSIVE.

Extracts individual plain-language observations and a 2-3 sentence expert summary.

Implementation: imageVerificationService.js (Lines 70-98)

6 Graceful Fallback on API Rate Limits

If Gemini Vision is temporarily unavailable or hits quota limits (HTTP 429/503), the system triggers buildMetadataOnlyImageResult, producing a heuristic forensic verdict based on EXIF tags, software signatures, and filename signals rather than failing.

Implementation: imageVerificationService.js (Lines 172-248)

4. URL Analysis Pipeline (Web & In-Browser Page Scanning)

Two distinct operational modes: direct URL submission and live browser DOM inspection

SatyaScan provides two complementary URL analysis workflows tailored for standalone web analysis and real-time extension browsing.

Mode A: Direct Web App URL Submission (POST /api/analyze)

1 Fast HTTP Scraping & HTML Sanitization

Fetches URL content using Axios with standard browser headers. Strips scripts, stylesheets, navigation bars, ads, and footers. Extracts semantic <article> or <main> containers and decodes HTML entities into clean text up to 10,000 characters.

Implementation: `urlScrapeService.js` (`extractTextFromUrl`)

2 Heuristic URL Type Classification

Classifies URLs into 5 distinct categories before analysis via domain sets, TLD suffixes, and path patterns:

- official: Government (.gov, .nic.in), international agencies (who.int, un.org), central banks (rbi.org.in).
- reference: Wikipedia, Britannica, Encyclopedia, Khan Academy.
- opinion: Editorial paths (/opinion/, /editorial/, /blogs/, /op-ed/, Substack, Medium).
- social_media: Twitter/X, Reddit, Facebook, Instagram, YouTube, Threads, Telegram.
- news: Standard news publications.

Implementation: `urlClassifierService.js` (`classifyUrl`)

3 Category-Tailored Verification Prompts

Non-news pages (official and reference) use `buildUrlTypePrompt` instead of standard true/false news prompts. This prevents authoritative documents (like WHO health advisories or government gazettes) from being falsely labeled as unverified news claims.

Implementation: `prompts/urlTypePrompts.js`, `textVerificationService.js`

Mode B: Chrome Extension Live Page Scanning (POST /api/analyze/page)

1 Content Script In-Page DOM Extraction

The extension content script walks the active DOM, filters hidden or non-content elements, extracts the article headline (<h1>), page title, OpenGraph description, and page body up to 12,000 characters.

Implementation: `satyascan-extension/src/content/index.js`

2 Two-Stage Claim Extraction & Full Article Fact-Check

- Stage 1 (Claim Extraction): POST /api/analyze/page/extract isolates the primary asserting claim, secondary claims, and key entities from the article body.
- Stage 2 (Page Verification): POST /api/analyze/page fact-checks the main claim against Google Fact Check and Tavily, analyzes political bias (Left, Center, Right), identifies suspicious/exaggerated statements, and highlights missing context.

Implementation: `services/pageAnalysisService.js`, `prompts/pageAnalysis.js`

5. System Resilience, Localization & Component Matrix

Architectural guarantees, multi-language support, and component responsibility mapping

Resilience & Performance Engineering

- **Retry Mechanism with Exponential Backoff:** withRetry wrapper in geminiService.js automatically recovers from transient network glitches (500, 502, 503, 504 timeouts).
- **API Quota Classification:** classifyGeminiError detects 429 quota exhaustion and returns graceful evidence-only results without crashing.
- **Sub-SS Execution:** Parallel retrieval across Google Fact Check and Tavily minimizes latency.
- **Full Bilingual Architecture:** End-to-end support for English and Hindi (Devanagari script), automatically switching prompts and output schemas based on user preference.

Component Responsibility & Code Matrix

MODULE / AREA	PRIMARY SOURCE FILE	RESPONSIBILITY & ARCHITECTURAL ROLE
Routing & Cache	analyzeController.js	Routes text/url/image, 24h cache, history logging
Text & URL Pipeline	textVerificationService.js	NER, query generation, evidence ranking, Gemini reasoning
Image Forensics	imageVerificationService.js	EXIF parsing, metadata integrity, Gemini Vision
Live Page Scanner	pageAnalysisService.js	Article extraction, main claim isolation, bias & context
Search Integration	tavilyService.js	Dual-pass search (broad + trusted domains), trust tiers
Fact-Check API	factCheckService.js	Google Fact Check Tools API query & normalization
URL Classifier	urlClassifierService.js	Heuristic classification: official, news, opinion, etc.
EXIF Parser	exifParser.js	Extracts camera, timestamps, GPS, ISO, aperture via exifr
Gemini Engine	geminiService.js	Google Generative AI client with retry, JSON parsing
Extension Worker	background/index.js	Manifest V3 Service worker, context menus, storage
Extension Content	content/index.js	DOM text walker, selection listener, auth sync

